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C++ Online Compiler

main.cpp

```
1 #include <iostream>
2 using namespace std;
3
4 char board[3][3] = {
5     {'1', '2', '3'},
6     {'4', '5', '6'},
7     {'7', '8', '9'}
8 };
9
10 char currentPlayer = 'X';
11
12 void displayBoard() {
13     cout << "\n";
14     cout << " " << board[0][0] << " | " << board[0][1] << " | " << board[0][2] << "\n";
15     cout << " " << board[1][0] << " | " << board[1][1] << " | " << board[1][2] << "\n";
16     cout << " " << board[2][0] << " | " << board[2][1] << " | " << board[2][2] << "\n";
17     cout << "-----\n";
18     cout << " " << board[0][0] << " | " << board[0][1] << " | " << board[0][2] << "\n";
19     cout << "\n";
20 }
21
22 bool makeMove(int choice) {
23     int row = (choice - 1) / 3;
24     int col = (choice - 1) % 3;
25
26     if (board[row][col] != 'X' && board[row][col] != 'O') {
27         board[row][col] = currentPlayer;
28         return true;
29     }
30     return false;
31 }
32
33 bool checkWin() {
34     // Rows
35     for (int i = 0; i < 3; i++) {
36         if (board[i][0] == currentPlayer &&
37             board[i][1] == currentPlayer &&
38             board[i][2] == currentPlayer) {
39             return true;
40         }
41     }
42     // Columns
43     for (int j = 0; j < 3; j++) {
44         if (board[0][j] == currentPlayer &&
45             board[1][j] == currentPlayer &&
46             board[2][j] == currentPlayer) {
47             return true;
48         }
49     }
50     // Diagonals
51     if (board[0][0] == currentPlayer &&
52         board[1][1] == currentPlayer &&
53         board[2][2] == currentPlayer) {
54         return true;
55     }
56     if (board[0][2] == currentPlayer &&
57         board[1][1] == currentPlayer &&
58         board[2][0] == currentPlayer) {
59         return true;
60     }
61     return false;
62 }
63
64 bool checkDraw() {
65     for (int i = 0; i < 3; i++) {
66         for (int j = 0; j < 3; j++) {
67             if (board[i][j] != 'X' && board[i][j] != 'O') {
68                 return false;
69             }
70         }
71     }
72     return true;
73 }
74
75 void switchPlayer() {
76     if (currentPlayer == 'X')
77         currentPlayer = 'O';
78     else
79         currentPlayer = 'X';
80 }
```

Output

```
1 | 2 | 3
---|---
4 | 5 | 6
---|---
7 | 8 | 9

Player X, enter position (1-9): 1
X | 2 | 3
---|---
4 | 5 | 6
---|---
7 | 8 | 9

Player O, enter position (1-9): 4
X | 2 | 3
---|---
0 | 5 | 6
---|---
7 | 8 | 9

Player X, enter position (1-9): 6
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 8 | 9

Player O, enter position (1-9): 8
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 0 | 9

Player X, enter position (1-9): 5
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 0 | 9
```

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main.cpp

```
43 }
44
45 // Columns
46 for (int i = 0; i < 3; i++) {
47     if (board[0][i] == currentPlayer &&
48         board[1][i] == currentPlayer &&
49         board[2][i] == currentPlayer) {
50         return true;
51     }
52 }
53
54 // Diagonals
55 if (board[0][0] == currentPlayer &&
56     board[1][1] == currentPlayer &&
57     board[2][2] == currentPlayer) {
58     return true;
59 }
60
61 if (board[0][2] == currentPlayer &&
62     board[1][1] == currentPlayer &&
63     board[2][0] == currentPlayer) {
64     return true;
65 }
66
67 return false;
68 }
69
70 bool checkDraw() {
71     for (int i = 0; i < 3; i++) {
72         for (int j = 0; j < 3; j++) {
73             if (board[i][j] != 'X' && board[i][j] != 'O') {
74                 return false;
75             }
76         }
77     }
78     return true;
79 }
80
81 void switchPlayer() {
82     if (currentPlayer == 'X')
83         currentPlayer = 'O';
84     else
85         currentPlayer = 'X';
86 }
```

Output

```
1 | 2 | 3
---|---
4 | 5 | 6
---|---
7 | 8 | 9

Player X, enter position (1-9): 1
X | 2 | 3
---|---
4 | 5 | 6
---|---
7 | 8 | 9

Player O, enter position (1-9): 4
X | 2 | 3
---|---
0 | 5 | 6
---|---
7 | 8 | 9

Player X, enter position (1-9): 6
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 8 | 9

Player O, enter position (1-9): 8
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 0 | 9

Player X, enter position (1-9): 5
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 0 | 9
```

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main.cpp

```
84 else
85     currentPlayer = 'X';
86 }
87
88 void resetBoard() {
89     char value = ' ';
90
91     for (int i = 0; i < 3; i++) {
92         for (int j = 0; j < 3; j++) {
93             board[i][j] = value;
94         }
95     }
96
97     currentPlayer = 'X';
98 }
99
100 int main() {
101
102     char replay;
103
104     do {
105         resetBoard();
106         bool gameOver = false;
107
108         while (!gameOver) {
109             displayBoard();
110
111             int choice;
112             cout << "Player " << currentPlayer << ", enter position (1-9): ";
113             cin >> choice;
114
115             if (choice < 1 || choice > 9) {
116                 cout << "Invalid position! Try again.\n";
117                 continue;
118             }
119
120             if (!makeMove(choice)) {
121                 cout << "Position already taken! Try again.\n";
122                 continue;
123             }
124
125             if (checkWin()) {
126                 displayBoard();
127                 cout << "Player " << currentPlayer << " wins!\n";
128                 gameOver = true;
129             }
130             else if (checkDraw()) {
131                 displayBoard();
132                 cout << "Game Draw!\n";
133                 gameOver = true;
134             }
135             else {
136                 switchPlayer();
137             }
138         }
139
140         cout << "Do you want to play again? (y/n): ";
141         cin >> replay;
142     } while (replay == 'y' || replay == 'Y');
143
144     cout << "Thank you for playing!\n";
145     return 0;
146 }
```

Output

```
1 | 2 | 3
---|---
4 | 5 | 6
---|---
7 | 8 | 9

Player X, enter position (1-9): 1
X | 2 | 3
---|---
4 | 5 | 6
---|---
7 | 8 | 9

Player O, enter position (1-9): 4
X | 2 | 3
---|---
0 | 5 | 6
---|---
7 | 8 | 9

Player X, enter position (1-9): 6
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 8 | 9

Player O, enter position (1-9): 8
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 0 | 9

Player X, enter position (1-9): 5
X | 2 | 3
---|---
0 | X | X
---|---
7 | 0 | 9

Player O, enter position (1-9): 7
X | 2 | 3
---|---
0 | X | X
---|---
0 | 0 | X

Player X, enter position (1-9): 9
X | 2 | 3
---|---
0 | X | X
---|---
0 | 0 | X

Player X wins!
Do you want to play again? (y/n):
```

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```
107
108 while (!gameOver) {
109     displayBoard();
110
111     int choice;
112     cout << "Player " << currentPlayer << ", enter position (1-9): ";
113     cin >> choice;
114
115     if (choice < 1 || choice > 9) {
116         cout << "Invalid position! Try again.\n";
117         continue;
118     }
119
120     if (!makeMove(choice)) {
121         cout << "Position already taken! Try again.\n";
122         continue;
123     }
124
125     if (checkWin()) {
126         displayBoard();
127         cout << "Player " << currentPlayer << " wins!\n";
128         gameOver = true;
129     }
130     else if (checkDraw()) {
131         displayBoard();
132         cout << "Game Draw!\n";
133         gameOver = true;
134     }
135     else {
136         switchPlayer();
137     }
138 }
139
140 cout << "Do you want to play again? (y/n): ";
141 cin >> replay;
142 } while (replay == 'y' || replay == 'Y');
143
144 cout << "Thank you for playing!\n";
145 return 0;
146 }
```

Output

```
Player X, enter position (1-9): 6
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 8 | 9

Player O, enter position (1-9): 8
X | 2 | 3
---|---
0 | 5 | X
---|---
7 | 0 | 9

Player X, enter position (1-9): 5
X | 2 | 3
---|---
0 | X | X
---|---
7 | 0 | 9

Player O, enter position (1-9): 7
X | 2 | 3
---|---
0 | X | X
---|---
0 | 0 | X

Player X, enter position (1-9): 9
X | 2 | 3
---|---
0 | X | X
---|---
0 | 0 | X

Player X wins!
Do you want to play again? (y/n):
```